Mid-Term Report

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1 Introduction

Solar and wind energies are intermittent sources, which constitue a caveat in electricity supply at national level.

Complementarity between wind tubines and solar panels (PVs) production could consistute an answer to intermittence. Indeed, it would insure a continuity in electricity production preventing for any fall in electricity supply.

In this work, we investigate complementarity through raw data using wind speed and solar radiation.

We use MeteoSwiss data, an open metorological database gathering weather related measures at 10-minutes slices, daily, monthly and yearly levels in most stations across Switzerland.

To assess complementarity between eletricity production from wind turbines and PVs, we chose to rely on mean wind speed in meter per second and mean solar radiation in watt per square meters. These measures are the natural sources by related renewables technologies.

The first measures of mean wind speed were made in 1864 in only 4 stations while, in 1980, 72 stations measured mean wind speed, reaching 146 stations in 2025. Regardin mean solar radiation, measures started in 1981 in 45 stations, reaching 129 stations in 2025. We exploit data on 1980-2025 range using the same stations across time for consistency reasons, despite a lower coverage compare to 2015-2025 period analysis which allows us to use 124 stations (measuring both solar radiation and wind speed) at country level.

1 shows the average wind mean wind speed and mean solar radiation over the 2015-2025 period. On purpose we kept the maximum of observation available in 2015 for each measures.

We aim here at studying the complementarity between wind speed and solar radiation to account for a potential electricity production complementarity between wind turbines and PVs. We will use correlations criteria as used by Canales et al. [2019] and product descriptive statistics allowing us

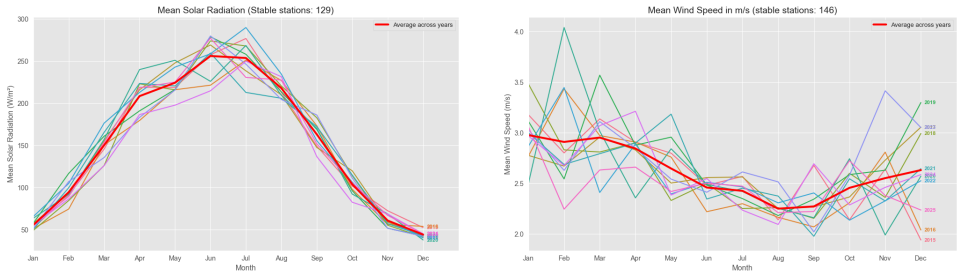


Figure 1: Mean wind speed and solar radiation from 2015 to 2025.

to analyse such complementarity. These statistics will cover a seasonal analysis, data stratification accounting for elevation and geographic zone using daily and monthly data.

References

Fausto A. Canales, Jakub Jurasz, Alexandre Beluco, and Alexander Kies. Assessing temporal complementarity between three variable energy sources by means of correlation and compromise programming. *arXiv*, 2019. doi: 10.1016/j.energy.2019.116637.